Module 1 Homework 4 Challenge

Provide the best answer for each of the following questions.

Time Remaining 00:59:08

## Main Question Set

Solve six of the first ten questions to achieve a passing score of 60%. Don't miss the learning opportunity in the rest of the problems, though -- spend the full time and attempt every one of them!

## 1

Review Later

Who gets more work done, $2x$ people who each work at a rate of $\frac{1}{2}\frac{\text{job}}{\text{hour}}$, or $x$ people who each work at a rate of $1\frac{\text{job}}{\text{hour}}$?

- $2x$ people who work at a rate of $\frac{1}{2}\frac{\text{job}}{\text{hour}}$
- $x$ people who work at a rate of $1\frac{\text{job}}{\text{hour}}$
- They get the same amount of work done in the same amount of time
- Not enough information to tell

## 2

Review Later

$140$ pounds of fresh cherries can be dried to get $21$ pounds of dried cherries. How many pounds of dried cherries can be gotten from $160$ pounds of fresh cherries?

- $18$
- $20$
- $22$
- $24$
- $21$

## 3

Review Later

What would be the rate of water flow of an intake pipe that can fill a swimming pool in $3$ hours?

- $1$ gallon per hour
- $\frac{1}{3}\frac{\text{pool}}{\text{hour}}$
- $\frac{1\text{ hour}}{\frac{1}{4}\text{ pool}}$

- $100$ gallons per foot
- $100$ gallons per cubic meter

## 4

Review Later

Which of the following is FALSE about the relationship between amount of work done, number of people doing the
work and the amount of time
spent doing the work?

- The bigger the number of people doing the work, the less time it takes
- The bigger the number of people doing the work, the more work that can be done
- The less time you have, the more people you need to do the work
- The less work that is done, the more time it takes
- The more time you have, the fewer the people that you need

## 5

Review Later

A company wants to make $250$ protractors and $60$ calculators every hour. A given worker can either make $10$ protractors or $4$ calculators per hour. How many workers will the company need to make the exact amount of products every hour?

## 6

Review Later

A large water tank has two inlet pipes (a large and a small one) and one outlet pipe. It takes $2$ hours to fill the tank with the large inlet pipe. On the other hand, it takes $5$ hours to fill the tank with the small inlet pipe. The outlet pipe allows the full tank to be emptied in $7$ hours. Assuming that the tank is initially empty, what fraction will be filled in $1$ hour if all three pipes are in operation? Your answer should be a fraction in simplest form, without spaces, e.g. 1/2.

## 7

Review Later

Alice and Bob start from the same position on a track and both start jogging clockwise along it. Alice runs at $4$ miles per hour, and Bob runs at $3$ miles per hour. How much extra distance will Alice have covered when she passes Bob?

- $1$ mile
- $2$ miles
- One track lap
- $\frac{3}{4}$ of a track lap
- $\frac{4}{3}$ of a track lap

## 8

Review Later

You just finished a triathlon in a spectacular $2$ hours, and you want to know how fast you swam. The race consists of $1$ mile of swimming, $12$ miles of bicycling, and $3$ miles of running. You know that you biked for $1$ hour and you ran at a speed of $5$ miles per hour. How many miles per hour did you swim? Your answer should be a number only, rounded to the nearest tenth, without units.

## 9

Review Later

The harmonic mean, a type of average of two speeds, is different from simply the arithmetic average of two rates, because

- If you go equal distance at speed number one and the same distance at speed number two, you don't spend equal amounts of time at each speed, since the faster speed will get you there in less time
- A mean is different from an average
- We need to find the median instead of the mean

- An arithmetic mean is the sum of a set of numbers divided by the number of numbers in the set
- The harmonic mean tells us the middle value of two speeds

## 10

Review Later

If $a$ ants can dig $b$ tunnels in $c$ hours, how many tunnels can $b$ ants dig in $a$ hours?

- $\frac{b^2}{c}$
- $\frac{a^2}{b}$
- $\frac{c^2}{a}$

## Bonus Challenge Questions

These optional questions will not count towards your final score. However, we strongly suggest trying them for the benefit of the challenge!

## 11

Review Later

A train travels at a constant speed of $120$ meters per second. The front of the train has just reached the entrance of a tunnel. Two seconds later, all of the train is inside the tunnel, and $3$ seconds after that, the back of the train exits the tunnel. What is the length of the train divided by the length of the tunnel? Your answer should be a fraction in simplest form, without spaces, e.g. 1/2

## 12

Review Later

A cyclist travels at a constant speed for $6$ hours over a distance $d$. How much longer, in hours, will it take for him to travel if he goes $50\%$ longer distance? Your answer should be the number only, without units.

## 13

Review Later

The long hand of a clock takes $60$ minutes for one rotation, while the small hand takes $12$ hours for one rotation. If it is $2:00$ p.m. now, at what time will the two hands overlap for the $5^{\text{th}}$ time?

- 5:25 p.m.
- 5:27 p.m.
- 6:30 p.m.
- 6:33 p.m.

## 14

Review Later

A car travels at a speed of $15$ kilometers per hour (kph) for $4$ hours, then travels for a distance of $30$ kilometers at a speed of $20$ kph, then $144$ kilometers at $36$ kph, and $12$ kph for the last

$30$ minutes. What was the kilometers per hour of the car's average speed?

## 15

Review Later

A car is driving at $40$ mph. It passes a train station. Fifteen minutes later, a train passes through the station at $60$ mph. How far from the train station will the car be when it is passed by the train, if both vehicles stay at a constant speed the whole time? Your answer should be the number only, without units.

## 16

Review Later

Elizabeth leaves home at $7:30$ p.m., walking at $3$ miles per hour. Once she has gone $4$ miles, Robert leaves the same house and bikes on her path at $19$ miles per hour. How many minutes later after $7:30$ p.m. do Elizabeth and Robert meet? Your answer should be the number only, without units.

## 17

Review Later

Suppose you take a trip with your car and drive uphill for $40$ miles, which uses up $\frac{1}{6}$ of your gas tank, and then you drive your car downhill for $60$ miles, which uses up another $\frac{1}{6}$ of your tank. You arrive at Numberland and have a day of math fun. Before you leave to go home, you fill your tank with $2$ gallons of gas, and then you make the trip back using the same route. Assuming that your tank has a capacity of $12$ gallons and you left home with a full tank, how many gallons of gas will you have when you get back home? Your answer should be a decimal, rounded to the nearest tenth, without units.

## Super Bonus Questions

These questions are an extreme challenge!  Some of the most interesting, creative and clever questions are the hardest ones. So try every single one. Even if you can't figure it out, you'll still be deepening your understanding of the important math concepts that underlie this course.

## 18

Review Later

A pirate tries to explode an enemy ship by lighting a fuse in the ammunition hold. Every minute, $\frac{1}{4}^{\text{th}}$ of the fuse burns away. He spends $16$ seconds running through the ship, which is $10$ meters long, and then escapes into a jolly-boat which he rows to shore $20$ meters away. His rowing speed is one-fifth of his running speed, and if he brings $n$ treasure chests onto the jolly-boat, his rowing speed reduces by $\frac{n}{10+n}$ of his original rowing speed. How many treasure chests can he carry to shore in one trip before the fuse detonates the explosives?

## 19

Review Later

Train stations $A$ and $B$ are connected by a straight track that is $72$ miles long. A slow freight train traveling at $18$ miles per hour leaves $A$ and heads towards $B$. At the same time, a bird who flies at $30$ miles per hour leaves $B$ and heads towards $A$. When the bird meets the train, it turns around "instantaneously" (that is, imagine it gets reflected) and heads back towards $B$. When the bird reaches the station, it turns around and heads back towards $A$ until it meets the train. Then the bird then turns around again and heads back towards the station. This process continues until the train reaches $B$. What is the total distance flown by the bird?

## 20

Review Later

Each sheet of paper costs $\$0.0025$ and each cardboard cover costs $\$0.20$. If a notebook is made using $168$ sheets of paper and two cardboard covers, how many **dollars** does it cost to manufacture $200$ notebooks?

Your answer should be the number only, without units.

View Marked Questions

Finished

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